

1. What photosynthesis is

Photosynthesis is the process by which plants, algae and some bacteria convert light energy into chemical energy stored in glucose. Organisms that do this are called autotrophs because they make their own food. The overall balanced equation is $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. In plants the process happens inside chloroplasts, which are most abundant in the mesophyll cells of leaves. Chloroplasts contain the green pigment chlorophyll, which absorbs mainly red and blue light and reflects green light, which is why leaves look green. Carbon dioxide enters the leaf through small pores called stomata, and water arrives from the roots through the xylem. Photosynthesis has two stages: the light-dependent reactions and the Calvin cycle.

2. The light-dependent reactions

The light-dependent reactions take place in the thylakoid membranes of the chloroplast. Light is absorbed first by photosystem II and then by photosystem I. In photosystem II, light energy is used to split water molecules in a step called photolysis. Splitting water releases electrons, hydrogen ions and oxygen gas, so the oxygen released by plants comes from water, not from carbon dioxide. The energised electrons pass along an electron transport chain, and the energy they release pumps hydrogen ions into the thylakoid space. The ions flow back through the enzyme ATP synthase, which makes ATP. This is called chemiosmosis. At the end of the chain, photosystem I passes electrons to NADP^+ , forming NADPH. ATP and NADPH then power the Calvin cycle.

3. The Calvin cycle

The Calvin cycle, also called the light-independent reactions, takes place in the stroma, the fluid that surrounds the thylakoids. It has three phases: carbon fixation, reduction and regeneration. In carbon fixation the enzyme RuBisCO attaches one molecule of carbon dioxide to a five-carbon sugar called RuBP. The unstable six-carbon product immediately splits into two molecules of 3-PGA. In the reduction phase, ATP and NADPH from the light-dependent reactions convert 3-PGA into G3P, a three-carbon sugar. In the regeneration phase most of the G3P is used, with more ATP, to rebuild RuBP so the cycle can continue. Making one G3P molecule that leaves the cycle requires three turns, fixing three carbon dioxide molecules and using nine ATP and six NADPH. Two G3P molecules can be combined to form one glucose.

4. Factors that limit photosynthesis

The rate of photosynthesis depends on light intensity, carbon dioxide concentration and temperature. The factor in shortest supply is called the limiting factor, because raising any other factor will not increase the rate. Increasing light intensity raises the rate until it levels off at the light saturation point. Temperature matters because the Calvin cycle is driven by enzymes; above roughly 40 degrees Celsius these enzymes begin to denature and the rate falls sharply. In hot, dry conditions stomata close to save water, carbon dioxide runs low and RuBisCO starts binding oxygen instead, a wasteful process called photorespiration. C4 plants such as maize reduce photorespiration by first fixing carbon dioxide into a four-carbon compound. CAM plants such as cacti open their stomata only at night, storing carbon dioxide as an acid and so reducing water loss during the day.

5. Aerobic cellular respiration

Cellular respiration releases the energy stored in glucose and captures it as ATP. The overall equation is $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$. It has three main stages. Glycolysis happens in the cytoplasm and does not need oxygen: one glucose is split into two pyruvate molecules, with a net gain of two ATP and two NADH. Pyruvate then enters the mitochondrion and is converted to acetyl-CoA. The Krebs cycle, in the mitochondrial matrix, produces two ATP, six NADH, two FADH₂ and four carbon dioxide molecules per glucose. Finally, the electron transport chain on the inner mitochondrial membrane uses NADH and FADH₂ to make most of the ATP by oxidative phosphorylation, with oxygen as the final electron acceptor, forming water. In total one glucose yields about 30 to 32 ATP.

6. Fermentation and how the two processes connect

When oxygen is not available, cells can keep making a little ATP through fermentation.

Fermentation follows glycolysis and its main purpose is to regenerate NAD^+ so that glycolysis can continue. It yields only the two ATP made in glycolysis, which is why it is far less efficient than aerobic respiration. In lactic acid fermentation, which happens in human muscle cells during intense exercise, pyruvate is converted into lactate. In alcoholic fermentation, carried out by yeast, pyruvate is converted into ethanol and carbon dioxide; this is used in brewing and in making bread rise. Photosynthesis and respiration are complementary: the glucose and oxygen produced by photosynthesis are the reactants of respiration, and the carbon dioxide and water released by respiration are the reactants of photosynthesis.